

The rank-one map $\ell^1 \rightarrow \ell^\infty$ in arXiv:2603.01336 is disjoint weak-star Dunford-Pettis

Source passage

After Definition 3.2, Boumnidel and Hafidi consider the operator

$$T : \ell^1 \longrightarrow \ell^\infty, \quad T((\lambda_k)_{k=1}^\infty) = \left(\sum_{k=1}^\infty \lambda_k, \sum_{k=1}^\infty \lambda_k, \dots \right),$$

and state that this T is not disjoint weak-star Dunford-Pettis. The relevant source pages are reproduced in Figures 1 and 2.

Claim

Theorem 1. *Let $T : \ell^1 \rightarrow \ell^\infty$ be defined by*

$$Tx = \left(\sum_{k=1}^\infty x_k \right) \mathbf{1}, \quad \mathbf{1} = (1, 1, \dots).$$

Then T is disjoint weak-star Dunford-Pettis.

Proof. Let (x_n) be a norm-bounded disjoint sequence in ℓ^1 , and let $(f_n) \subset (\ell^\infty)'$ satisfy $f_n \rightarrow 0$ in the weak-star topology $\sigma((\ell^\infty)', \ell^\infty)$. Put

$$\sigma(x) = \sum_{k=1}^\infty x_k, \quad x \in \ell^1.$$

Then $\sigma \in (\ell^1)'$, so the scalar sequence $(\sigma(x_n))$ is bounded. Also, since $\mathbf{1} \in \ell^\infty$ and $f_n \rightarrow 0$ weak-star,

$$f_n(\mathbf{1}) \longrightarrow 0.$$

Therefore

$$f_n(Tx_n) = f_n(\sigma(x_n)\mathbf{1}) = \sigma(x_n)f_n(\mathbf{1}) \longrightarrow 0.$$

This is precisely the definition of a disjoint weak-star Dunford-Pettis operator. □

Remark 1. *The displayed calculation in the source cannot produce a counterexample: any weak-star null sequence in $(\ell^\infty)'$ must vanish on the constant vector $\mathbf{1}$. In particular, coordinate functionals e_n are not weak-star null in $(\ell^\infty)'$, since $e_n(\mathbf{1}) = 1$ for every n . The argument above only refutes the printed assertion about this particular rank-one operator; it does not address whether some different example can serve the intended purpose.*

Since $(\pi_{k_n} \circ T)$ is w^* -null sequence, by [[8], Lemma 2.2] we have (f_n) is w^* -null sequence in E' . By hypothesis, it follows from Definition 3.1 that $f_n(x_n) \rightarrow 0$ as $n \rightarrow \infty$. On the other hand,

$$|f_n(x_n)| = |(\pi_{k_n} \circ T)(x_n)| = |\pi_{k_n}(T(x_n))| = T(x_n) > \epsilon > 0.$$

This is a contradiction. Hence, $\|T(x_n)\| \rightarrow 0$ for every disjoint sequence (x_n) of E , that is, T is an M-weakly compact operator.

(2) $\Rightarrow$ (1) By Definition 3.1, to prove that E has the disjoint weak* Dunford-Pettis property, we need to prove that for each norm bounded disjoint sequence (x_n) of E and each disjoint w^* -null sequence $(f_n) \subset E'$, we have $f_n(x_n) \rightarrow 0$. Let consider the operator $T : E \rightarrow c_0$ defined by $T(x) = (f_n(x))_n$ for any $x \in E$. By hypothesis, T is M-weakly compact. Therefore, $\|T(x_n)\| \rightarrow 0$ as $n \rightarrow \infty$, hence $f_n(x_n) \rightarrow 0$, as desired.

(2) $\Leftrightarrow$ (3) By virtue of [[17], Theorem 18]. □

Proposition 3.3. *Let E be a Banach lattice with the disjoint weak* Dunford-Pettis* property, If E' and E are order continuous, then E is finite dimensional.*

Proof. Let (x_n) be a disjoint norm bounded sequence of E , by Lemma 2 and Lemma 1 of [16], $|x_n| \xrightarrow{uaw} 0$, then by [[16], Theorem 7] we have $|x_n| \xrightarrow{w} 0$. (*)

Let (f_n) be a norm bounded disjoint sequence in E' , by [10, Corollary 2.4.3], f_n is w^* -null sequence, by assumption, $f_n(x_n) \rightarrow 0$. (**)

From (*) and (**) and [[4], Corollary 2.6], we conclude that $x_n \xrightarrow{\|\cdot\|} 0$.

Now, by [[17], Theorem 18] and [[11], Corollary 5] we claim that E is finite dimensional. □

We introduce the weak* version of uaw-Dunford-Pettis operators.

Definition 3.2. *An operator $T : E \rightarrow Y$ from a Banach lattice E into a Banach space Y is said to be disjoint weak* Dunford-Pettis, if for every norm bounded disjoint sequence (x_n) in E , and every sequence $f_n \xrightarrow{w^*} 0$ in Y' imply, $\lim_{n \rightarrow \infty} f_n(T(x_n)) = 0$.*

Now, we consider some ideal properties for the class of disjoint weak* Dunford-Pettis operator. But before investigating this, we give the following Example.

$$\begin{aligned} T : \ell^1 &\longrightarrow \ell^\infty \\ \lambda_n &\longmapsto \left(\sum_{n=1}^{\infty} \lambda_n, \sum_{n=1}^{\infty} \lambda_n, \dots \right) \end{aligned}$$

Figure 1: Definition 3.2 and the displayed rank-one operator.

By Remark 3.3 and Proposition 3.7 Id_{ℓ^∞} is a disjoint weak* Dunford-Pettis operator, However $T = Id_{\ell^\infty} \circ T$ is not disjoint weak* Dunford-Pettis. Indeed, let $f_n \xrightarrow{w^*} 0$ in $(\ell^\infty)'$, then there exists a unique sequence of scalars (α_n) such that $f_n = \sum_{n=1}^{\infty} \alpha_n e_n$. Moreover, we have that

$$\begin{aligned} \langle T(e_n), f_n \rangle &= \langle (\sum_{n=1}^{\infty} e_n, \sum_{n=1}^{\infty} e_n, \dots), \sum_{n=1}^{\infty} \alpha_n e_n \rangle \\ &= \sum_{n=1}^{\infty} \alpha_n \langle (\sum_{n=1}^{\infty} e_n, \sum_{n=1}^{\infty} e_n, \dots), e_n \rangle \\ &= \sum_{n=1}^{\infty} \alpha_n \cdot \sum_{n=1}^{\infty} e_n \langle (1, 1, \dots), e_n \rangle \\ &= \sum_{n=1}^{\infty} \alpha_n \cdot \sum_{n=1}^{\infty} e_n \rightarrow 0 \end{aligned}$$

We clearly see that T is not a disjoint weak* Dunford-Pettis operator. We also claim that T is not disjointness preserving operator. Indeed, since (e_n) is disjoint then $T(e_n \wedge e_m)$ is null in ℓ^∞ , but

$$T(e_n) \wedge T(e_m) = (1, 1, 1, \dots) \neq (0, 0, 0, \dots)$$

Proposition 3.4. *Let $S : E \rightarrow F$ and $T : F \rightarrow G$ be two operators between Banach lattices E , F and G .*

- (1) *If T is bounded and S is a disjoint weak* Dunford-Pettis operator, then $T \circ S$ is also disjoint weak* Dunford-Pettis.*
- (2) *If S is disjointness preserving and T is a disjoint weak* Dunford-Pettis operator, then $T \circ S$ is disjoint weak* Dunford-Pettis.*

Proof. (1) Let (x_n) be a norm bounded disjoint sequence of E and $(f_n) \subset G'$ such that $f_n \xrightarrow{w^*} 0$. Then $f_n \circ T \xrightarrow{w^*} 0$ $(\star)$.

By assumption S is disjoint weak* Dunford-Pettis, it follows from $(\star)$ that $f_n \circ T(S(x_n)) = f_n(T \circ S(x_n)) \rightarrow 0$, for each null weak* convergent sequence (f_n) in G' . Hence $T \circ S$ is disjoint weak* Dunford-Pettis.

(2) Let (x_n) be a disjoint sequence of E and $(f_n) \subset G'$ such that $f_n \xrightarrow{w^*} 0$. As, S is disjointness preserving operator, it follows that $(S(x_n))$ is a disjoint sequence. In this step, assumption asserts that $f_n(T(S(x_n))) \rightarrow 0$. Therefore, $T \circ S$ is disjoint weak* Dunford-Pettis. $\square$

Figure 2: The source passage claiming that T is not disjoint weak-star Dunford-Pettis.